THE COTANGENT COMPLEX OF AN ALGEBRAIC SPACE OVER A RING IS THE COTANGENT COMPLEX OF SHEAVES OF RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the identification and names the constant sheaf; the proof is a three-step reduction ending in a vanishing statement for strict henselizations.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The distinguished triangle for a composition of morphisms of ringed topoi. Lemma lemma-triangle-ringed-topoi, used at proof lines 4531-4533. Reduces the statement to the vanishing of L_q.
- Exactness of the stalk functor and the description of the stalk of the cotangent complex. Properties of Spaces, Theorem spaces-properties-theorem-exactness-stalks and Lemma lemma-stalk-cotangent-complex, used at proof lines 4535-4542. Turn the vanishing into a statement about one local ring.
- The vanishing of the cotangent complex of a strict henselization over the ring. Lemma lemma-when-zero, used at proof lines 4543-4547. Gives the required vanishing.
- The etale local ring as a strict henselization of a local ring. Properties of Spaces, Lemma spaces-properties-lemma-describe-etale-local-ring, used at proof lines 4543-4546. Identifies the ring appearing in the stalk computation.

Source reference. Original lemma 08T0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11148–11148. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08V1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11219–11219. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08R2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11080–11080. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04KF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5971–5971. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original theorem 04K5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5961–5961. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let Λ be a ring. Let X be an algebraic space over Λ. Then*

$$L_{X/\operatorname{Spec}(\Lambda)} = L_{\mathcal{O}_X/\underline{\Lambda}}$$

where $\underline{\Lambda}$ is the constant sheaf with value Λ on $X_{étale}$.

Proof. Let $p : X \to \operatorname{Spec}(\Lambda)$ be the structure morphism. Let $q : \operatorname{Spec}(\Lambda)_{étale} \to (*, \Lambda)$ be the obvious morphism. By the distinguished triangle of Lemma 08V1 it suffices to show that $L_q = 0$. To see this it suffices to show (Properties of Spaces, Theorem 04K5) for a geometric point $\bar{t} : \operatorname{Spec}(k) \to \operatorname{Spec}(\Lambda)$ that

$$(L_q)_{\bar{t}} = L_{\mathcal{O}_{\operatorname{Spec}(\Lambda)_{étale},\bar{t}}/\Lambda}$$

(Lemma 08T0) is zero. Since $\mathcal{O}_{\operatorname{Spec}(\Lambda)_{étale},\bar{t}}$ is a strict henselization of a local ring of Λ (Properties of Spaces, Lemma 04KF) this follows from Lemma 08R2. □